blood glucose levels. Since glucagon reduces the cellular uptake and utilisation of glucose it is called a hyperglycemic hormone. Prolonged hyperglycemia leads to the disorder called diabetes mellitus.

**Humulin N: Human insulin is produced by recombinant DNA technology (genetic engineering) and administered to diabetic patients as injection and not by oral consumption. Reason: Digestive enzymes digest it.**




**Insulin:** The half life period of insulin (in plasma) is 6 minutes

It is cleared from the circulation within 10-15 minutes

Endocrine glands control and coordinate the body functions through secreting certain chemical messengers called hormones. Due to certain physiological reasons, the blood glucose level of an otherwise normal person.

a. Give the possible cause for the increases in blood glucose level.

b. What is the chemical nature of this hormone? Discuss its role in the body.

c. How can this condition be reversed?

## 11.2.9 Gonads

**Testis:** A pair of testis is present in the scrotal sac of males. The testis functions as a sex organ and also as an endocrine gland. The testis is composed of seminiferous tubules and interstitial cells or Leydig cells. The Leydig cells secrete several male sex hormones, collectively called androgens, mainly testosterone.

**Functions of testosterone:** Under the influence of FSH and LH, testosterone initiates maturation of male reproductive organs, and the appearance of secondary sexual characters, muscular growth, growth of facial and axillary hair, masculine voice and male sexual behaviour. It enhances the total bone matrix and plays a stimulating role in the process of spermatogenesis.

**Ovary:** Females have a pair of ovaries located in the pelvic region of the abdomen. The ovary is composed of ovarian follicles and stromal tissues. It produces the eggs or ova. The ovaries secrete the steroid hormones oestrogen and progesterone. **Oestrogen** is responsible for the maturation of reproductive organs and the development of secondary sexual characters at puberty. Along with progesterone, oestrogens promotes breast development and initiate the cyclic changes during menstrual cycle. **Progesterone** prepares the uterus for implantation of the fertilized ovum. It decreases the uterine contraction during pregnancy and stimulates the development of mammary glands and milk secretion. It is responsible for premenstrual changes in the uterus and is essential for the formation of placenta.

Identify the peaks of FSH, LH, Oestrogen and Progesterone hormones through out the menstrual cycle.

Urine pregnancy test is done to test the presence of Human Chorionic Gonadotrophin (HCG) in the urine. HCG can be detected in the urine one or two weeks after conception.



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